

Mehdi Karbalayghareh, Ph.D.

CONTACT INFORMATION

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PROFESSIONAL EXPERIENCE

Purdue University West Lafayette, IN
Postdoctoral Research Associate May 2025 – present
Working with Dr. [David J. Love](#) and Dr. [Christopher G. Brinton](#) at the intersection of artificial intelligence/machine learning (AI/ML), wireless communications, and distributed optimization, with applications in next-generation wireless networks (6G and beyond) and distributed intelligence.

Purdue University West Lafayette, IN
Lecturer of Electrical and Computer Engineering Jan. 2026 – May 2026
Instructor for ECE 302 – Probabilistic Methods for Electrical and Computer Engineering.

Nokia Bell Labs Murray Hill, NJ
Network Systems Researcher Jun. 2024 – Apr. 2025
Worked with Drs. Dora van Veen, Vincent Houtsma, Jochen Maes, and Robert Borkowski on non-orthogonal multiple access (NOMA) schemes with interference-cancellation receivers for multi-user passive optical networks.

The University of Texas at Dallas Richardson, TX
Graduate Research Assistant Aug. 2019 – May 2024
Worked with Dr. [Aria Nosratinia](#) on developing novel NOMA transmission schemes for multi-user multiple-input multiple-output (MIMO), massive MIMO, and reconfigurable intelligent surface (RIS) systems under channel state feedback, where the links experience unequal mobility (fading speed) and/or non-identical spatial correlation conditions.

Ozyegin University Istanbul, Turkey
Graduate Research Assistant Sept. 2016 – July 2019
Worked with Dr. [Murat Uysal](#) on developing advanced communication systems for underwater and vehicular applications, focusing on 5G and beyond technologies.

EDUCATION

The University of Texas at Dallas Richardson, TX
Doctor of Philosophy (Ph.D.) in Electrical Engineering, *Advisor: Aria Nosratinia* 2019 – 2024

Ozyegin University Istanbul, Turkey
Master of Science (M.Sc.) in Electrical Engineering, *Advisor: Murat Uysal* 2016 – 2019

Iran University of Science and Technology Tehran, Iran
Bachelor of Science (B.Sc.) in Electrical Engineering 2011 – 2015

HONORS AND AWARDS

- **ECE Research Excellence Award**, The University of Texas at Dallas 2024
- **Excellence in Education Doctoral Fellowship**, The University of Texas at Dallas 2024
- **IEEE ComSoc Student Grant**, IEEE International Conference on Communications (ICC) 2024
- **Excellence in Education Doctoral Fellowship**, The University of Texas at Dallas 2023
- **Best Paper Award**, IEEE International Black Sea Conference on Communications and Networking (BlackSeaCom) 2019
- **IEEE Student Grant**, IEEE Wireless Communications and Networking Conference (WCNC) 2019
- **Full Ph.D. Scholarship**, The University of Texas at Dallas 2019–2024
- **Tubitak Fellowship**, The Scientific & Technological Research Council of Turkey 2016–2019
- **Full M.Sc. Scholarship**, Ozyegin University 2016–2019

RESEARCH INTERESTS

Machine learning; distributed optimization and edge learning; wireless communications and sensing; next-generation wireless networks (6G and beyond); information theory with applications to learning, communications, security, and privacy

PUBLICATIONS

Journal Papers

- J12 **M. Karbalayghareh**, A. Rajasekaran, X. Ma, D. J. Love, and C. G. Brinton, “AISAC: Closing the Loop Between AI and Integrated Sensing and Communication for 6G,” *submitted to IEEE Communications Magazine*, 2026.
- J11 **M. Karbalayghareh**, D. J. Love, and C. G. Brinton, “Learning OTA: A Unified Framework for Edge Sensing, Computation, and Communication,” *submitted to IEEE Transactions on Wireless Communications*, 2026.
- J10 A. Rajasekaran, **M. Karbalayghareh**, X. Ma, D. J. Love, and C. G. Brinton, “Optimal Multi-RIS Placement: Coverage-Guaranteed Sum Rate Maximization Under Inhomogeneous User Distributions,” *submitted to IEEE Transactions on Wireless Communications*, 2026.
- J9 **M. Karbalayghareh**, D. J. Love, and C. G. Brinton, “Coherence-aware over-the-air distributed learning under heterogeneous link impairments,” *IEEE Journal on Selected Areas in Information Theory*, 2026.
- J8 **M. Karbalayghareh** and A. Nosratinia, “Advanced Channel Training and Enhanced Capacity under Antenna Correlation in Multi-User RIS Systems,” *IEEE Transactions on Wireless Communications*, 2025.
- J7 **M. Karbalayghareh** and A. Nosratinia, “Pilot-Domain NOMA for RIS-Assisted Communications,” *IEEE Transactions on Communications*, 2024.
- J6 **M. Karbalayghareh** and A. Nosratinia, “Multi-User Pilot-Domain NOMA under Coherence Disparity and Channel State Feedback,” *IEEE Transactions on Wireless Communications*, 2024.
- J5 **M. Karbalayghareh**, F. Miramirkhani, H. B. Eldeeb, R. C. Kizilirmak, S. M. Sait, and M. Uysal, “Channel Modelling and Performance Limits of Vehicular Visible Light Communication Systems,” *IEEE Transactions on Vehicular Technology*, 2021.
- J4 M. Uysal, S. Ghasvarianjahromi, **M. Karbalayghareh**, P. D. Diamantoulakis, G. K. Karagiannidis, and S. M. Sait, “SLIPT for Underwater Visible Light Communications: Performance Analysis and Optimization,” *IEEE Transactions on Wireless Communications*, 2021.
- J3 F. Miramirkhani, **M. Karbalayghareh**, E. Zeydan, and R. Mitra, “Enabling 5G Indoor Services for Residential Environment using VLC Technology,” *Physical Communication*, 2022.
- J2 F. Miramirkhani, **M. Karbalayghareh**, and R. Mitra, “Least Minimum Symbol Error Rate based Post-Distortion for Adaptive Mobile VLC Transmission with Receiver Selection,” *Physical Communication*, 2022.
- J1 F. Miramirkhani, **M. Karbalayghareh**, and M. Uysal, “Effect of Scattering Phase Function on Underwater Visible Light Communication Channel Models,” *Physical Communication*, 2021.

Conference Papers

- C13 **M. Karbalayghareh**, D. J. Love, and C. G. Brinton, “Learning-Aware Rate Splitting for Decentralized Federated Learning over Wireless Graphs,” *submitted to IEEE INFOCOM*, 2027.
- C12 **M. Karbalayghareh**, D. J. Love, and C. G. Brinton, “Distributed Edge Learning under Imperfect Data Sensing,” *accepted for publication in IEEE ITW*, 2026.
- C11 **M. Karbalayghareh**, D. J. Love, and C. G. Brinton, “Coherence-Aware Distributed Learning under Heterogeneous Downlink Impairments,” *IEEE INFOCOM*, 2026.
- C10 A. Rajasekaran, **M. Karbalayghareh**, X. Ma, D. J. Love, and C. G. Brinton, “Optimal RIS Placement in Multi-User MISO Systems with User Randomness,” *IEEE WiOpt*, 2026.
- C9 **M. Karbalayghareh** and A. Nosratinia, “Pilot-Domain NOMA for Multi-User RIS-Assisted Communications,” *IEEE ICC*, 2024.
- C8 **M. Karbalayghareh** and A. Nosratinia, “RIS-Assisted Downlink Transmission under Unequal Coherence Intervals and CSI Feedback,” *IEEE PIMRC*, 2023.

- C7 **M. Karbalayghareh** and A. Nosratinia, “Interaction of Pilot Reuse and Channel State Feedback under Coherence Disparity,” *IEEE ITW*, 2022.
- C6 **M. Karbalayghareh**, F. Miramirkhani, M. Safari, and M. Uysal, “Vehicular Visible Light Communications with SPAD Receivers,” *IEEE WCNC*, 2019.
- C5 S. Ghasvarianjahromi, **M. Karbalayghareh**, P. D. Diamantoulakis, G. K. Karagiannidis, and M. Uysal, “Simultaneous Lightwave Information and Power Transfer in Underwater Visible Light Communications,” *IEEE PIMRC*, 2019.
- C4 M. Elamassie, **M. Karbalayghareh**, F. Miramirkhani, M. Uysal, M. Abdallah, and K. Qaraqe, “Resource Allocation for Downlink OFDMA in Underwater Visible Light Communications,” *IEEE BlackSeaCom*, 2019.
- C3 A. Zeshan, **M. Karbalayghareh**, F. Miramirkhani, M. Uysal, and T. Baykas, “Comparative Performance Evaluation of VLC, LTE and WLAN Technologies in Indoor Environments,” *IEEE BlackSeaCom*, 2021.
- C2 M. Elamassie, **M. Karbalayghareh**, F. Miramirkhani, and M. Uysal, “Adaptive DCO-OFDM for Underwater Visible Light Communications,” *IEEE SIU*, 2019.
- C1 M. Elamassie, **M. Karbalayghareh**, F. Miramirkhani, R. C. Kizilirmak, and M. Uysal, “Effect of Fog and Rain on the Performance of Vehicular Visible Light Communications,” *IEEE VTC*, 2018.

RESEARCH PROPOSALS

1. **NSF VINES Track 2: Heterogeneous Edge Intelligence for Natural Disaster Response** [\$4.8M - awarded]
Analytics (HEIDRA)
 - Role: Senior Personnel — lead contributor to the thrust on heterogeneous distributed intelligence.
 - PIs: C. G. Brinton, D. J. Love, and S. Sundaram (Purdue University); Y. Sun (Auburn University); C. Shen and J. Yang (University of Virginia).
 - Industry Partners: Nokia and Saab, Inc.
2. **DARPA Ag × BTO: Actionable Latent Space of Crop Signatures (SignAI)** [\$0.9M - awarded]
 - Role: contributor to multimodal sensing and generative AI modeling of crop signatures.
 - PIs: C. G. Brinton, D. J. Love, and J. Krogmeier (Purdue University).
3. **NSF NeTS: Distributed Intelligence in Heterogeneous Wireless Networks** [under review]
 - Role: contributor to the thrust on predictive wireless modeling and scheduling, including adaptive CSI acquisition, channel prediction, and communication design under heterogeneous channel dynamics.
 - PIs: C. G. Brinton and D. J. Love (Purdue University).

SELECTED PROJECTS

AI-integrated sensing and communication (AISAC) for 6G edge intelligence

- Proposed a closed-loop AISAC framework that unifies AI, sensing, and communication, enabling physical-world observations to support edge training, inference, and network adaptation.
- Developed the learning-alignment principle for jointly configuring waveforms, beams, bandwidth, transmit power, and sensing modes based on downstream learning utility rather than sensing accuracy or communication rate alone.
- Characterized the three-way sensing–communication–learning tradeoff under imperfect sensing and developed a vehicular edge-intelligence use case for learning-aware physical-layer design.

Distributed AI/ML over wireless networks (6G applications)

- Developed coherence-aware federated-learning frameworks that account for heterogeneous channel dynamics (unequal coherence times), enabling joint scheduling of dynamic and static devices.
- Integrated over-the-air aggregation with coherence-aware scheduling, accounting for channel-estimation errors (downlink and uplink) and partial model updates by dynamic devices.
- Developed adaptive clustering and resource-allocation algorithms leveraging mobility, channel state, and data heterogeneity to improve convergence under non-i.i.d. conditions.

Coherence and correlation diversity in multi-user wireless systems assisted by RIS

- Developed innovative NOMA schemes for multi-user MIMO systems under unequal link coherence conditions and channel-state feedback.
- Developed pilot-domain NOMA in RIS-assisted MIMO communications, significantly reducing channel-training overhead while achieving gains in rate and degrees of freedom.

- Exploited correlation diversity by combining pilot reuse in common eigenspaces and rate-splitting in non-common eigenspaces in RIS-assisted MIMO communications.
- Developed joint optimization of transmit beamforming and RIS reflection coefficients under coherence and correlation disparities.

TEACHING EXPERIENCE

- **ECE 440 – Transmission of Information** (Guest Lecturer), Purdue University Sept. 2026
- **ECE 302 – Probabilistic Methods for Electrical and Computer Engineering** (Lecturer), Purdue University Spring 2026
- **Convex Optimization** (Teaching Assistant), The University of Texas at Dallas Spring 2024
- **Signal Theory** (Teaching Assistant), The University of Texas at Dallas Fall 2023
- **Information Theory** (Teaching Assistant), The University of Texas at Dallas Spring 2023
- **Probability and Statistics** (Teaching Assistant), Ozyegin University 2018–2019
- **Linear Algebra** (Teaching Assistant), Ozyegin University 2017–2018
- **Mathematics for Engineering** (Teaching Assistant), Ozyegin University 2016–2017

STUDENT MENTORING

- **David Nickel** (Ph.D. student), Purdue University Jan. 2026 – present
- **Xiaoyan Ma** (Ph.D. student), Purdue University Jun. 2025 – present
- **Abhishek Rajasekaran** (Ph.D. student), Purdue University Jun. 2025 – present

TALKS AND CONFERENCE PRESENTATIONS

- IEEE Information Theory Workshop (ITW), Tempe, AZ, talk (upcoming) Nov. 2026
- IEEE International Conference on Computer Communications (INFOCOM), Tokyo, Japan, talk May 2026
- Information Theory and Applications Workshop (ITA), San Diego, CA, poster Feb. 2026
- IEEE International Conference on Communications (ICC), Denver, CO, talk Jun. 2024
- Information Theory and Applications Workshop (ITA), San Diego, CA, talk Feb. 2024
- ECE Research Day, The University of Texas at Dallas, Richardson, TX, talk Nov. 2023
- IEEE Int. Symp. on Personal, Indoor and Mobile Radio Comm. (PIMRC), Toronto, Canada, talk Sept. 2023
- Information Theory and Applications Workshop (ITA), San Diego, CA, poster Feb. 2023
- IEEE Information Theory Workshop (ITW), Mumbai, India, virtual Nov. 2022
- Information Theory and Applications Workshop (ITA), San Diego, CA, poster Feb. 2020
- IEEE Int. Symp. on Personal, Indoor and Mobile Radio Comm. (PIMRC), Istanbul, Turkey, talk Sept. 2019
- IEEE Wireless Communications and Networking Conference (WCNC), Marrakesh, Morocco, talk Apr. 2019
- IEEE Vehicular Technology Conference (VTC), Porto, Portugal, talk Jun. 2018

PROFESSIONAL SERVICE

Technical Program Committee

- IEEE International Conference on Communications (ICC) 2027

Session Chair

- IEEE International Conference on Computer Communications (INFOCOM), two technical sessions 2026

Reviewer

- IEEE Communications Magazine, IEEE Vehicular Technology Magazine, IEEE Transactions on Wireless Communications (TWC), IEEE Transactions on Communications (TCOM), IEEE/ACM Transactions on Networking (TON), IEEE Transactions on Vehicular Technology (TVT), IEEE Transactions on Cognitive Communications and Networking (TCCN), IEEE Journal of Lightwave Technology (JLT), IEEE Communications Letters (COMML), IEEE Wireless Communications Letters (WCL), IEEE International Conference on Communications (ICC), IEEE Global Communications Conference (GLOBECOM), IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), IEEE Vehicular Technology Conference (VTC), IEEE National Conference on Communications (NCC), IEEE Military Communications Conference (MILCOM), Physical Communication (Elsevier), ACM International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc)